



# Python for Fixed Income

COURSE 2 · BOND MATH

## Duration and DV01

WEEK 2 · DAY 4

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# From price to price risk

If yields move 1 bp, what does each position make or lose?

| Issuer          | Par held  | Dirty price |
|-----------------|-----------|-------------|
| Quorvane Energy | 5,000,000 | 100.236551  |
| Dunmarrow Rail  | 3,000,000 | 101.010586  |
| Pellwick Paper  | 4,000,000 | 101.179914  |
| Halvering Foods | 2,000,000 | 99.872491   |

Settled Monday 30 November 2026, at day 8's prices. The positions are invented.

## Week 2: dates, accrued, and risk

**Day 6** Dates and coupon schedules

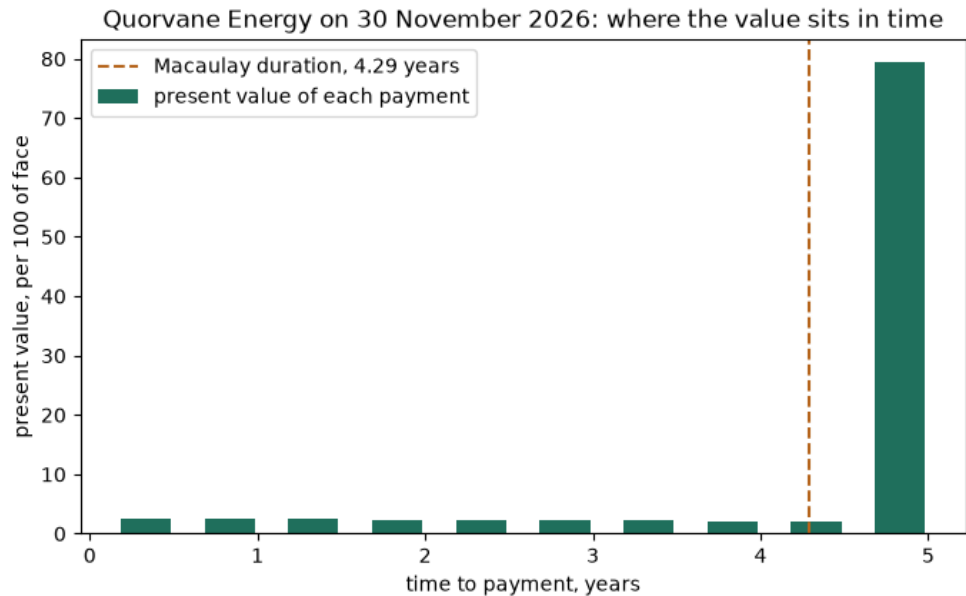
**Day 7** Day counts and accrued interest

**Day 8** Clean and dirty price

**Day 9** **Duration and DV01**

**Day 10** Convexity, the September book

# Macauley duration: the balance point in time



Each payment's time, weighted by its present value over the dirty price.

## 4.2932

years: Quorvane's Macaulay  
duration

4.8333

years to the final payment

*Macaulay, NBER, 1938: duration is "a reality of which 'maturity' is only one factor".*

### Excel side by side (tested in Excel)

=DURATION(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,2,0)

4.293248

=DURATION(DATE(2026,11,30),DATE(2031,9,30),0,5.4/100,2,0)

4.833333

## Modified duration: price sensitivity

```
modified = Macaulay / (1 + y / 2)
```

Typed out, the division returns MDURATION to the last digit. The /2 is the frequency.

The price falls about modified duration percent for each 100 bp rise in yield.

| Quorvane, dirty  | Repriced    | Duration line |
|------------------|-------------|---------------|
| +1 bp, percent   | -0.04179333 | -0.04180378   |
| +100 bp, per 100 | -4.087492   | -4.190266     |
| -100 bp, per 100 | +4.296930   | +4.190266     |

## Big moves: above the line, both ways.

Duration is the slope at today's yield, and the curve bends. Day 10 measures the bend: convexity.

### Excel side by side (tested in Excel)

=MDURATION(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,2,0)

4.180378

=DURATION(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,2,0)/(1+5.4/100/2)

4.180378

# DV01 per 100 and per position

$$\text{DV01} = \text{modified} \times \text{dirty price} \times 0.0001$$

| Issuer          | DV01 per 100 | Position, dollars |
|-----------------|--------------|-------------------|
| Quorvane Energy | 0.041903     | 2,095.13          |
| Dunmarrow Rail  | 0.055175     | 1,655.26          |
| Pellwick Paper  | 0.026359     | 1,054.35          |
| Halvering Foods | 0.067511     | 1,350.23          |
| Book            |              | 6,154.97          |

Position DV01 in dollars = DV01 x par held / 100. Dollars add up across positions.

Excel side by side (tested in Excel)

**B1**      =PRICE(...)

**B2**      =100\*5.25/100/2\*COUPDAYBS/COUPDAYS

**B4**      =MDURATION(...)

**B5**      =B4\*(B1+B2)/10000

**B11**     =B5\*5000000/100

**B5 returns 0.041903; B11 returns 2,095.13.**

# Check DV01 by bumping the yield

```
down = bondmath.price(s, m, 5.25, 5.4 - 0.01)
up = bondmath.price(s, m, 5.25, 5.4 + 0.01)
(down - up) / 2      # half the 2 bp change
```

Clean prices are fine: accrued does not move with yield, so it cancels in the difference.

**Not for a bond in its final period: Excel's PRICE uses simple interest there.**

## Excel side by side (tested in Excel)

|     |                                |           |
|-----|--------------------------------|-----------|
| B8  | =PRICE(...,(5.4-0.01)/100,...) | 99.403465 |
| B9  | =PRICE(...,(5.4+0.01)/100,...) | 99.319659 |
| B10 | =(B8-B9)/2                     | 0.041903  |

**1.9e-9**

per 100: B10 minus the DV01 formula in B5

# Signs: what a positive DV01 means

Change in value for a move of  $n$  bp = minus position DV01 x  $n$

**Long: 5,000,000 Quorvane**

Position DV01: +2,095.13 dollars

Yield 1 bp lower: gains about that much

Yield 1 bp higher: loses about that much

**Short: -2,000,000 Halvering**

Position DV01: -1,350.23 dollars

Yield 1 bp lower: loses about that much

Yield 1 bp higher: gains about that much

**This course reports DV01 positive for a long. Some systems flip it: check before adding two sources.**

Arithmetic on signs, not a view on where yields go.

# The holdings duration column, reproduced

**200**

bonds in holdings.csv, as of 30 September 2026

**0.456824**

years: Macaulay's largest miss against the file

**0.000498**

years: modified duration's largest gap, within rounding

**The file's duration column is modified duration: price sensitivity.**

```
holdings["modified_python"] = [bondmath.modified_duration(*bond) for bond in inputs]
assert holdings["gap"].max() <= 0.0005 + 1e-9    # the file rounds to 3 decimals
```

# Today's Excel, every formula tested

| Job          | Excel (tested), Quorvane                                          | Returns  |
|--------------|-------------------------------------------------------------------|----------|
| Macaulay     | =DURATION(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,2,0)  | 4.293248 |
| Zero coupon  | =DURATION(DATE(2026,11,30),DATE(2031,9,30),0,5.4/100,2,0)         | 4.833333 |
| Modified     | =MDURATION(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,2,0) | 4.180378 |
| DV01 per 100 | =B4*(B1+B2)/10000                                                 | 0.041903 |
| Position     | =B5*5000000/100                                                   | 2,095.13 |
| The bump     | =(B8-B9)/2                                                        | 0.041903 |

**B1 PRICE, B2 the accrued formula, B4 MDURATION; B8 and B9 PRICE 1 bp either side.**

# Look back: a file's history

**git log --oneline -- bondmath.py**

```
ff99a3a Add macaulay_duration, ...  
957fc63 Start of day 9: ...
```

Only the commits that touched bondmath.py. Everything after -- is a file name.

**git diff HEAD~1 -- bondmath.py**

```
+def macaulay_duration(settlement: ...  
+def modified_duration(settlement: ...  
+def dv01(settlement: date, ...
```

HEAD is the last commit, HEAD~1 the one before. Every line is +: added, nothing taken out.

**A test that passed yesterday fails today? The diff since yesterday is the first place to look.**

Hashes differ on every run.

# Bring it home: paste, test, commit, look back

1

## Your folder

Set-Location to my\_bondmath, in your projects folder.

2

## Paste

Three functions and four tests at the end of your files, without the %%writefile line.

3

## Test

python -m pytest -q: 27 passed, or 35 with the exercises of days 1 to 8.

4

## Commit

git diff --stat, git add, git commit with a subject and a body.

5

## Look back

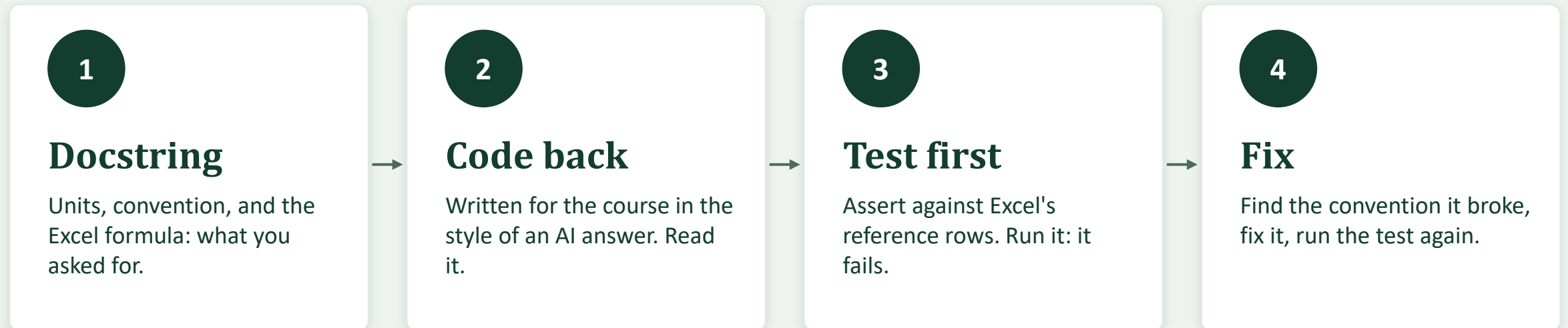
git log --oneline -- bondmath.py: a commit for every day you brought home.

```
git diff HEAD~1 -- bondmath.py
```

In Colab: download bondmath.py and test\_bondmath.py from the work folder under /tmp. They replace yesterday's downloads.

# The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.



Today: modified duration divides by  $(1 + y)$ , the whole annual yield, instead of  $(1 + y / 2)$ .

*Code that runs is not the same as code that is right.*

# What you can do now

- Compute Macaulay duration and match Excel's DURATION
- Turn it into modified duration and match MDURATION
- Compute DV01 per 100 and per position, and a book DV01
- Check DV01 by bumping the yield; read its sign, long and short
- Reproduce the holdings duration column; look back with git log and git diff

**Week 2, Day 5** Convexity, where duration's straight line misses, and the September book